ECS 334: Linear Control Systems

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1 Abstract

This report studies a new distributed adaptive continuous-time convex optimization algorithm over weight-balanced directed and strongly connceted graphs with adaptive constants. The study can be further extended to weight-unbalanced settings and event-triggered communication among agents. We present the problem formulation, key analytical results, and equilibrium characterization within the convergence analysis. Simulation results are provided to demonstrate the algorithm's behavior and to validate the theoretical findings. These types of algorithms are particularly useful for optimization involving non-convex global cost functions, unknown network connectivity, and nonlinear system dynamics.

2 Introduction

In recent decades, distributed optimization has attracted significant attention due to its wide range of applications, in operations of swarm of robots, sensor networks, resource allocation, and distributed learning. A central problem in this area involves minimizing a global objective function expressed as the sum of local cost functions, where agents rely only on local computation and communication with neighboring nodes. While a large body of work has focused on discrete-time, consensus-based algorithms, continuous-time approaches have emerged as an attractive alternative due to their analytical tractability and the ability to leverage classical stability tools for convergence analysis.

Existing continuous-time distributed optimization algorithms have achieved notable progress, particularly over undirected and weight-balanced directed graphs [3] [2]. However, these methods rely on restrictive assumptions such as global knowledge of network parameters and strong convexity of gradients. These requirements limit their applicability in large-scale or uncertain networks, where connectivity information may be unavailable and objective functions may exhibit nonlinear [1] and/or non-convex behavior [4]. Moreover, the extension to directed communication topologies introduces additional challenges due to asymmetric information flow and coupled system dynamics.

3 Notation

Let $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{R}^{n\times m}$ denote the sets of real numbers, real n-dimensional vectors, and real $n \times m$ matrices, respectively. Let $\mathbb{R}_{>0}$ denote the set of positive real numbers. The transpose of a vector or matrix is denoted by $(\cdot)^T$. The identity matrix of dimension n is denoted by I_n, and the all-ones vector in $\mathbb{R}^n$ is denoted by $\mathbf{1}_n$.

For a vector $x \in \mathbb{R}^n$, $\|x\|$ denotes its Euclidean norm. The ith eigenvalue of a matrix A is denoted by $\lambda_i(A)$. The symbol $\otimes$ denotes the Kronecker product.

For vectors $u_1, \ldots, u_m$, we write $u = [u_1, \ldots, u_m]$ to denote the aggregated vector. In a networked system, local variables associated with agent i are denoted using a superscript; for example, x^i denotes the state of agent i. If $p^i \in \mathbb{R}^d$ is a variable of agent i, then the aggregated (along columns) variable of the N-agent network is written as

$$p = [p^1, \ldots, p^N] \in (\mathbb{R}^d)^N.$$

For convenience, we use $\mathbf{r} \in \mathbb{R}^N$ and $\mathbf{R} \in \mathbb{R}^{N\times(N-1)}$, defined by

$$\mathbf{r} = \frac{1}{\sqrt{N}}\mathbf{1}_N, \qquad \mathbf{r}^T\mathbf{R} = 0, \qquad \mathbf{R}^T\mathbf{R} = I_{N-1}. \tag{3.1}$$

A directed graph is denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, where $\mathcal{V} = \{1, 2, \ldots, n\}$ is the set of agents, $\mathcal{E}$ is the set of directed edges, and $A = [a_{ij}] \in \mathbb{R}^{n\times n}$ is the weighted adjacency matrix. The Laplacian matrix of the graph is denoted by $L = D - A$, where $D = \mathrm{diag}(d_1, \ldots, d_n)$ is the out-degree matrix with $d_i = \sum_{j=1}^n a_{ij}$.

$$\text{A digraph } G \text{ is weight-balanced} \quad \Longleftrightarrow \quad \big(\mathbf{1}_N^T L = 0\big) \quad \Longleftrightarrow \quad \big(\forall x \in \mathbb{R}^N,\ x^T L x \geq 0\big)$$

Each agent $i \in \mathcal{V}$ is associated with a local cost function $f_i : \mathbb{R}^p \to \mathbb{R}$, and the global optimization problem is to minimize the aggregate cost

$$f(x) = \sum_{i=1}^n f_i(x).$$

The gradient of f_i is denoted by $\nabla f_i(x)$, and the gradient of the global cost is denoted by $\nabla f(x)$.

A set $\Omega \subseteq \mathbb{R}^p$ is said to be convex if, for all $x, y \in \Omega$ and all $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in \Omega.$$

A differentiable function f is said to be strongly convex on a convex set $\Omega \subseteq \mathbb{R}^p$ if there exists $m > 0$ such that

$$(x - y)^T\big(\nabla f(x) - \nabla f(y)\big) \geq m\|x - y\|^2, \quad \forall x, y \in \Omega.$$

A function f is said to be locally Lipschitz if, for every $x \in \mathbb{R}^p$, there exists a neighborhood W of x and a constant $M \geq 0$ such that

$$\|f(y) - f(z)\| \leq M\|y - z\|, \quad \forall y, z \in W.$$

4 Problem Statement

The formulation in this section follows [4]. Consider a set of N agents interacting over a directed communication graph. Each agent $i \in \{1, 2, \ldots, N\}$ is associated with a local cost function

$f_i : \mathbb{R}^n \to \mathbb{R}$. The global cost function is defined as

$$f(z) = \sum_{i=1}^N f_i(z).$$

The objective is to design a continuous-time distributed algorithm such that each agent can solve the optimization problem

$$\min_{z\in\mathbb{R}^n} f(z) \tag{4.1}$$

using only its own information and information received from its neighboring agents.

4.1 Assumptions

The following assumptions are adopted throughout this report.

Assumption 1. The global cost function f is differentiable and strongly convex over $\mathbb{R}^n$. Each local cost function f_i is differentiable, and its gradient is locally Lipschitz on $\mathbb{R}^n$.

Under Assumption 1, the strong convexity of the global cost function guarantees that the optimization problem admits a unique solution.

Note that this assumption does not require each local cost function f_i to be convex. Instead, it suffices that their aggregation satisfies the strong convexity condition mentioned above, which can arise even if some of the individual f_i are nonconvex.

Assumption 2. The communication graph $\mathcal{G}$ is strongly connected.

Assumption 3. The initial condititons of the internal states of the agents sum to 0

4.2 Equivalent Distributed Formulation

Under Assumption 2, the optimization problem can be reformulated in a distributed form as

$$\min_{x_i \in \mathbb{R}^n} \tilde{F}(x) = \sum_{i=1}^{N} f_i(x_i), \quad \text{subject to } (L \otimes I_n)x = 0 \tag{4.2}$$

where x_i is the state of agent i, and

$$x = [x_1^T, x_2^T, \ldots, x_N^T]^T$$

denotes the network state.

The constraint $(L \otimes I_n)x = 0$ represents the consensus condition. In particular,

$$(L \otimes I_n)x = 0 \implies \exists \tau \in \mathbb{R}^n \text{ such that } x = \mathbf{1}_N \otimes \tau.$$

The proof of this equivalence follows the same argument as Lemma 3.1 in [2]. Therefore, all agents agree on a common decision variable, and the original optimization problem is equivalent to the distributed constrained formulation above. This reformulation makes the problem suitable for distributed algorithm design and analysis.

4.3 Project Objectives

The objectives of this project are as follows:

- Study the assumptions and conditions required for the convergence of the distributed optimization algorithm.
- Derive the equilibrium points and establish the convergence proof for the algorithm that solves the stated optimization problem.
- Provide simulation results to support the theoretical analysis.

Report Structure: Section 5 presents the derivation of the equilibrium points and the proof of convergence by considering an appropriate Lyapunov function and showing that the algorithm converges to the equilibrium points. Section 6 presents simulations for a few prominent directed graph topologies, and Section 7 provides an analysis of the simulation results together with concluding remarks.

5 Distributed Adaptive Continuous-Time Convex Optimization Algorithm on Weight-Balanced and Strongly Connected Directed Graphs

In this section, we present a fully distributed algorithm for solving the optimization problem in Equation (2). The algorithm is inspired by the adaptive continuous-time distributed optimization method proposed in [4], with modifications inspired by [3].

For each agent $i \in \{1, 2, \ldots, N\}$, consider the dynamics

$$\dot{v}^i = \gamma_1(\alpha_i + \beta_i) \sum_{j=1}^{N} a_{ij}(x^i - x^j), \tag{5.1}$$

$$\dot{x}^i = -\gamma_2 \nabla f_i(x^i) - \gamma_1(\alpha_i + \beta_i) \sum_{j=1}^{N} a_{ij}(x^i - x^j) - v^i, \tag{5.2}$$

$$\dot{\alpha}_i = \beta_i. \tag{5.3}$$

where

$$\beta_i = e_i^T e_i, \qquad e_i = \sum_{j=1}^{N} a_{ij}(x^i - x^j).$$

Here, $v^i \in \mathbb{R}^n$ denotes the internal state of agent i, e_i denotes the relative consensus error, and $\gamma_1, \gamma_2 \in \mathbb{R}_{>0}$ are control gains. The quantities α_i and β_i act as adaptive coupling gains, with $\alpha_i(0) > 0$.

The algorithm is distributed in the sense that each agent updates its state using only its own information and information received from its neighbors. In the modified form considered here, Equation (5.2) contains just the term v^i instead of

$$\sum_{j=1}^{N} a_{ij}(v^i - v^j).$$

As a result, neighboring agents are not required to exchange their internal states through the communication graph.

The overall network dynamics can be written compactly as

$$\dot{v} = \gamma_1 \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)x, \tag{5.4}$$

$$\dot{x} = -\gamma_2 \tilde{f}(x) - \gamma_1 \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)x - v, \tag{5.5}$$

$$\dot{\hat{\alpha}} = \hat{\beta}, \tag{5.6}$$

where

$$\hat{\alpha} = \mathrm{diag}(\alpha_1, \alpha_2, \ldots, \alpha_N), \qquad \hat{\beta} = \mathrm{diag}\left(\|e_1\|^2, \ldots, \|e_N\|^2\right),$$
$$e = (L \otimes I_n)x,$$
$$v = [(v^1)^T, (v^2)^T, \ldots, (v^N)^T]^T,$$

and

$$\tilde{f}(x) = [(\nabla f_1(x^1))^T, (\nabla f_2(x^2))^T, \ldots, (\nabla f_N(x^N))^T]^T.$$

This compact form is useful for analyzing the equilibrium points and convergence properties of the algorithm in the subsequent sections.

5.1 Equilibrium Points

To determine the equilibrium points, we set

$$\dot{x} = 0, \qquad \dot{v} = 0, \qquad \dot{\hat{\alpha}} = 0,$$

and denote an equilibrium by $(\bar{x}, \bar{v})$.

From (5.6), the condition $\dot{\hat{\alpha}} = 0$ implies $\hat{\beta} = 0$. Since

$$\hat{\beta} = \mathrm{diag}(\|e_1\|^2, \ldots, \|e_N\|^2),$$

we obtain $e_i = 0$ for all i, and therefore

$$(L \otimes I_n)\bar{x} = 0. \tag{5.7}$$

Hence $\bar{x} \in \ker(L \otimes I_n)$, which implies

$$\bar{x} = \mathbf{1}_N \otimes \tilde{\theta}$$

for some $\tilde{\theta} \in \mathbb{R}^n$.

Next, from the equilibrium condition (5.5), we have

$$\bar{v} = -\gamma_2 \nabla \tilde{f}(\bar{x}) \quad \Longleftrightarrow \quad \bar{v}^i = -\gamma_2 \nabla f_i(\bar{x}^i), \quad \forall i \in \{1, \ldots, N\}. \tag{5.8}$$

Summing over i and using Assumption 3 gives

$$\sum_{i=1}^{N} \bar{v}^i = 0 \quad \Longrightarrow \quad \sum_{i=1}^{N} \nabla f_i(\tilde{\theta}) = 0.$$

Equivalently,

$$\nabla \left(\sum_{i=1}^{N} f_i(\tilde{\theta}) \right) = 0. \tag{5.9}$$

By Assumption 1, the global cost function is strongly convex, so its stationary point is unique. Therefore,

$$\tilde{\theta} = x^* = \arg\min_x \sum_{i=1}^{N} f_i(x).$$

Substituting this into the consensus form of $\bar{x}$ yields

$$\bar{x} = \mathbf{1}_N \otimes x^*.$$

Thus, every equilibrium point is a consensus configuration at the global minimizer of (4.2). □

5.2 Proof of Convergence

Theorem 1. Under Assumptions 1–3, if the communication graph $\mathcal{G}$ is strongly connected and weight-balanced, then every trajectory of (5.1)–(5.3) converges to an equilibrium of the form

$$x = \mathbf{1}_N \otimes x^*,$$

where x^* is the unique minimizer of the global objective in (4.2). In particular, the algorithm solves the distributed optimization problem asymptotically, and each adaptive gain $\alpha_i(t)$ converges to a finite constant.

Proof: Let $(\bar{x}, \bar{v})$ be an equilibrium characterized in the previous subsection. Then

$$\bar{x} = \mathbf{1}_N \otimes x^*, \qquad \bar{v}^i = -\gamma_2 \nabla f_i(x^*), \quad i \in \{1, \dots, N\}.$$

Introduce the shifted variables

$$y = x - \bar{x}, \qquad u = v - \bar{v}. \tag{5.10}$$

Since $\bar{x}$ and $\bar{v}$ are constant, we have $\dot{y} = \dot{x}$ and $\dot{u} = \dot{v}$. Also, because $(L \otimes I_n)\bar{x} = 0$, the disagreement vector can be written as

$$e = (L \otimes I_n)x = (L \otimes I_n)y.$$

Define the stacked gradient deviation

$$h(y) := \tilde{f}(y + \bar{x}) - \tilde{f}(\bar{x}). \tag{5.11}$$

Using (5.4)–(5.6), the error dynamics become

$$\dot{u} = \gamma_1 \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)y, \tag{5.12}$$

$$\dot{y} = -\gamma_2 h(y) - \gamma_1 \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)y - u, \tag{5.13}$$

$$\dot{\hat{\alpha}} = \hat{\beta}. \tag{5.14}$$

The first step is to identify an invariance property. Because the graph is weight-balanced, $\mathbf{1}_N^T L = 0$. Hence,

$$\frac{d}{dt}(\mathbf{1}_N^T \otimes I_n)v = (\mathbf{1}_N^T \otimes I_n)\dot{v} = \gamma_1(\mathbf{1}_N^T L \otimes I_n)x = 0.$$

By Assumption 3, $(\mathbf{1}_N^T \otimes I_n)v(0) = 0$, so $(\mathbf{1}_N^T \otimes I_n)v(t) = 0$ for all $t \geq 0$. Therefore the internal-state constraint used in the equilibrium analysis is preserved along trajectories.

Next consider the nonnegative function

$$V(y, u) = \frac{1}{2}\|y\|^2 + \frac{1}{2}\|u\|^2. \tag{5.15}$$

Differentiating along (5.12)–(5.13) gives

$$\dot{V} = y^T \dot{y} + u^T \dot{u} \tag{5.16}$$

$$= -\gamma_2 y^T h(y) - \gamma_1 y^T \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)y - y^T u \tag{5.17}$$

$$+ \gamma_1 u^T \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)y. \tag{5.18}$$

By Assumption 1, the global cost is strongly convex, so there exists $m > 0$ such that

$$y^T h(y) \geq m\|y\|^2. \tag{5.19}$$

Moreover, because $\hat{\alpha}(t)$ is diagonal and nondecreasing with $\alpha_i(0) > 0$, the matrix $\hat{\alpha}(t) + \hat{\beta}(t)$ remains positive diagonal for all t. Since the graph is weight-balanced and strongly connected, the quadratic form

$$y^T \big[(\hat{\alpha} + \hat{\beta}) \otimes I_n\big](L \otimes I_n)y$$

is nonnegative and vanishes only on the consensus subspace. Therefore the second term in (5.18) penalizes disagreement.

The mixed terms in (5.18) can be bounded by Young's inequality. Consequently, for every sufficiently small $\varepsilon > 0$, there exists a constant $c > 0$ such that

$$\dot{V} \leq -(\gamma_2 m - \varepsilon)\|y\|^2 - c\|(L \otimes I_n)y\|^2 + \varepsilon\|u\|^2. \tag{5.20}$$

This estimate implies boundedness of y and u, and it also shows that the disagreement signal $(L \otimes I_n)y$ belongs to L_2. Since $\dot{\alpha}_i = \beta_i = \|e_i\|^2 \geq 0$, each adaptive gain is nondecreasing. Furthermore, because $e_i(t) \to 0$ as $t \to \infty$, the quantity $\beta_i(t)$ tends to zero, and each $\alpha_i(t)$ converges to a finite limit.

Now apply LaSalle's invariance principle to the largest invariant set contained in

$$\{(y, u) : \dot{V} = 0\}.$$

From (5.20), the identity $\dot{V} = 0$ implies $(L \otimes I_n)y = 0$, so y lies in the consensus subspace. Thus there exists $z \in \mathbb{R}^n$ such that

$$x = \mathbf{1}_N \otimes z.$$

On this manifold, the dynamics reduce to

$$\dot{z} = -\gamma_2 \sum_{i=1}^{N} \nabla f_i(z),$$

while the invariance property still gives $\sum_{i=1}^{N} v^i = 0$. Hence every invariant point in this set satisfies

$$\sum_{i=1}^{N} \nabla f_i(z) = 0.$$

By the strong convexity of the global objective, this condition has the unique solution $z = x^*$. Therefore the largest invariant set inside $\{\dot{V} = 0\}$ is precisely the equilibrium

$$x = \mathbf{1}_N \otimes x^*.$$

LaSalle's principle now yields

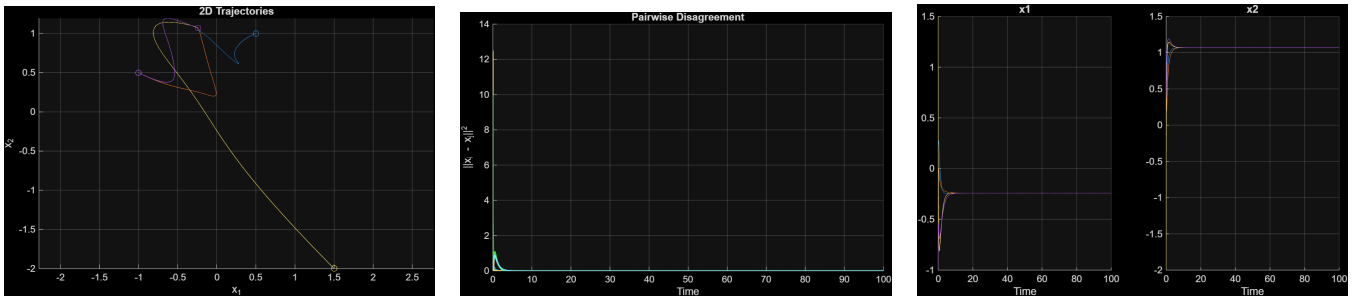
$$\lim_{t\to\infty} x(t) = \mathbf{1}_N \otimes x^*.$$

Hence the algorithm solves (4.2) expoenetially, and the adaptive gains converge to finite steady-state values. □

6 Simulations

The simulations were implemented in MATLAB; the source code is available here.

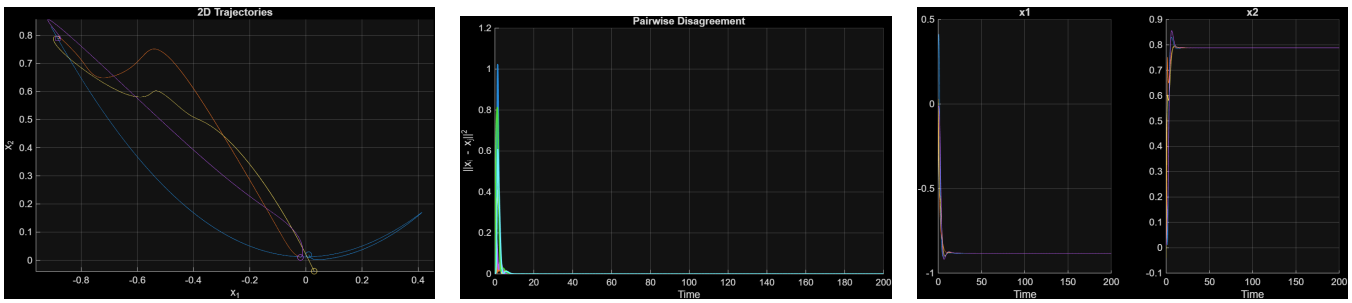
Simulation 1. Four agents with two-dimensional states interact over a square graph with heterogeneous local objectives. Using $T = 100$ and $dt = 0.001$, the states converge to a common optimizer.



(a) State trajectories (b) Pairwise disagreement (c) Evolution of x_1 and x_2

Figure 1: Simulation 1 results.

Simulation 2. Four agents with two-dimensional states interact over the same graph, but with nonlinear local objectives including Rosenbrock, double-well, sinusoidal, and coupled quadratic terms. Using $T = 200$ and $dt = 0.001$, the states still converge to a common consensus point.



(a) State trajectories (b) Pairwise disagreement (c) Evolution of x_1 and x_2

Figure 2: Simulation 2 results.

7 Conclusion

By the proof of convergence produced above the dynamics in (5.4)–(5.6) solves the distributed problem (4.2), and the simuations validate the observations.

References

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